

ROBERT CADDY

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EDUCATION

University of Pittsburgh, PA <i>Ph.D. Physics</i>	2018 - 2024
Bowling Green State University, OH <i>M.S. Physics</i>	2016 - 2018
Purdue University, IN <i>B.S. Honors Physics Major, Astronomy Minor</i>	2012 - 2016

TECHNICAL & PROFESSIONAL SKILLS

Programming Languages:	C++, C, Python, Fortran, Bash
Packages & APIs:	CUDA, HIP, OpenMP Offload, Kokkos, OpenMP, MPI, GoogleTest, Numpy, Polars
Software Tools:	Clang Tools, Make, HDF5, Slurm, Jenkins, Doxygen, Docker
HPC Systems Used:	OLCF: Frontier, Summit, Andes, Crusher. ALCF: Theta, Aurora. University Clusters

EXPERIENCE

Research Software Engineer II <i>Princeton University, Princeton, NJ</i>	2024 - Present
– Developed and optimized scientific codes in collaboration with Astrophysics research groups, specializing GPU accelerate codes in computational Astrophysics. – Worked with researchers to expand the capabilities of Iseult, a visualization code for Particle-in-Cell simulations. Added several key new features and updated the code to comply with best practices. – Performed a proof-of-concept port of the hydrodynamics code Ramses to GPUs. Demonstrated a 20-30x speedup, several times higher than initially expected. – Developed the PegasusTools python package for analyzing data from the Pegasus++ PIC code. I achieved 15-150x speedups in standard analysis workflows. – Optimized the multiphysics astrophysical simulation code Fornax to get 1.8x speedup on Aurora, an exascale supercomputer.	
Ph.D. Candidate <i>University of Pittsburgh, Pittsburgh, PA</i>	2018 - 2024
– Expanded Cholla, a massively parallel GPU-accelerated code for simulating astrophysical fluid dynamics to include magnetic fields (magnetohydrodynamics/MHD) using state of the art methods. – Collaborated with the Frontier Center for Accelerated Application Readiness (CAAR) program to optimize Cholla to run on exascale supercomputers, namely Frontier. – Established and executed a robust testing framework for Cholla, employing GoogleTest with custom extensions to ensure software reliability and quality. – Led multiple initiatives to promote scientific software best practices within the Cholla development team, fostering excellence in software engineering standards.	
Masters Student <i>Bowling Green State University, Bowling Green, OH</i>	2016 - 2018
– Conducted original thesis research into the properties of symbiotic star V1835 Aql, and other stars in the same field, with Professor Andrew Layden as advisor. – Determined the properties of variable star systems through image and data analysis in Python.	
Undergraduate Research Assistant <i>Purdue University, West Lafayette, IN</i>	2015 - 2016
– Built an experimental optics system to observe the effect of various chemotherapy drugs on cancer tumors. – Reduced experimental data analysis and archiving time from ~10 hours to ~30 minutes.	

HONORS & AWARDS

- **Spot Award**, Princeton University, 2025.
- **Learning Beyond the Classroom Certificate**, Purdue University, 2016.
 - Required work experience, volunteer time, career training, and one significant activity – bicycling across the U.S. with Bike & Build to raise money and awareness for affordable housing.
- **Presidential Scholarship**, Purdue University, 2012.
- **Ascarelli Fellow**, Purdue University Department of Physics and Astronomy, 2012.
- **Eagle Scout**, 2012. Project: Designed and built sheds designed to withstand ice falling from nearby building.

SERVICE

Tutorials

2021 - 2025

Developed and delivered tutorials ranging from 1 hour to half day.

- *Introduction to CUDA*, Princeton University, 2025
- *Introduction to Python Data Types*, University of Pittsburgh, 2023
- *Scientific Software Best Practices*, University of Pittsburgh 2022, 2023
- *Introduction to Git, GitHub, and Git Workflows*, University of Pittsburgh, 2022
- *Organizing Your Dotfiles*, University of Pittsburgh, 2021

Mentorship

2024 - 2025

- US-RSE'25 Conference Mentor, US Research Software Engineer Association, 2024
- Princeton Open Hackathon Mentor, Princeton University, 2025
- US-RSE'24 Conference Mentor, US Research Software Engineer Association, 2024

Teaching

2013 - 2020

- Graduate Teaching Assistant for introductory physics labs, University of Pittsburgh, 2018-2020
- Graduate Teaching Assistant for introductory physics labs, Bowling Green State University, 2016-2018
- Undergraduate Teaching Assistant for introductory physics courses, Purdue University, 2013-2016

President, Purdue Society of Physics Students (SPS)

2015 - 2016

Purdue University, West Lafayette, IN

- Tripled the number of active members and increased profits of the club store by ~5x.
- Coordinated biweekly events and activities such as designing and constructing a weather balloon payload with SPS National funding, inviting guest speakers for special colloquia, and visiting Argonne National Lab.

PUBLICATIONS & SELECTED TALKS

- *Dynamical Dark Energy Imprints in the Lyman-Alpha Forest*. Garza, D.; Robertson, B.; Madau, P.; Gnedin, N.; Abbruzzo, M.W.; Schneider, E.; Budiardja, R.D.; White, J.B., III; **Caddy, R.**; Villaseñor, B. 2026 DOI: [10.48550/arXiv.2601.00767](https://doi.org/10.48550/arXiv.2601.00767), arXiv:[2601.00767](https://arxiv.org/abs/2601.00767)
- **Caddy, R.** “Getting Scientist buy-in on best practices: A Case Study.” 2nd Annual Conference of the US Research Software Engineer Association (US-RSE'24), October 2024, Rapid Access Microtalk (RAM)
- *Cholla-MHD: An Exascale-capable Magnetohydrodynamic Extension to the Cholla Astrophysical Simulation Code*. **Caddy, R** & Schneider, E. 2024. The Astrophysical Journal, Volume 970 DOI: [10.3847/1538-4357/ad464a](https://doi.org/10.3847/1538-4357/ad464a), arXiv:[2402.05240](https://arxiv.org/abs/2402.05240)
- **Caddy, R.** “Exascale MHD Simulations with Cholla.” Santa Cruz Organization for Outreach in Physics (SCOOP), November 2023, University of California Santa Cruz. *Invited Talk*
- **Caddy, R.** “Exascale MHD Simulations with Cholla.” Seminar, November 2023, NASA Goddard. *Invited Talk*
- **Caddy, R.** “Exascale MHD Simulations with Cholla.” Center for Theory and Computation (CTC) Seminar, November 2023, University of Maryland. *Invited Talk*
- *Optical Time-series Photometry of the Symbiotic Nova V1835 Aquilae* **Caddy, R.**, Layden, A, et al. 2022. Publications of the Astronomical Society of the Pacific, Volume 134, Number 1039 DOI: [10.1088/1538-3873/ac8f6f](https://doi.org/10.1088/1538-3873/ac8f6f), arXiv:[2209.11251](https://arxiv.org/abs/2209.11251)

CONTINUING EDUCATION

- **3rd Annual Conference of the US Research Software Engineer Association (US-RSE'25)**, Conference 2025
- **ACM/IEEE Supercomputing (SC24)**, Conference 2024
- **2nd Annual Conference of the US Research Software Engineer Association (US-RSE'24)**, Conference 2024
- **ACM/IEEE Supercomputing (SC23)**, Conference 2023
- **1st Annual Conference of the US Research Software Engineer Association (US-RSE'23)**, Conference 2023
- **Practice and Experience in Advanced Research Computing (PEARC)**, Conference 2023
- **Platform for Advanced Scientific Computing (PASC)**, Conference 2023
- **Argonne Training Program for Extreme Scale Computing (ATPESC)** 2022
- **International High Performance Computing Summer School (IHPCSS)** 2022,
Second place in the programming challenge